• <u>Def</u> A Grassmann algebra is a vector space constructed from a set of n generators θ_i on which an antisymmetric product is defined:

$$\theta_i \theta_j + \theta_j \theta_i = 0$$

$* \ \theta_i \theta_i + \theta_i \theta_i = 2\theta_i\theta_i = 0 \implies \theta_i \theta_i = 0$

A generic element of a Grassmann algebra is therefore a 1-degree polynomial

$$f(\theta_i) = \sum_{k=0}^{n} \sum_{i_1, \dots, i_k}^{n} C^{(k)}_{i_1, \dots, i_k} \theta_{i_1} \theta_{i_2} \cdots \theta_{i_k}$$

• The dimension of the Grassmann algebra is then the number of distinct monomials that can be constructed from the θ_i, namely 2^n.

e.g. $(n=1)$ $\quad f(\theta) = C_0 + C_1 \theta$

$(n=2)$ $\quad f(\theta_1, \theta_2) = C_0 + C_1\theta_1 + C_2\theta_2 + C_{12}\theta_1\theta_2$

• The generators of a Grassmann algebra are often called Grassmann variables.

$f(\theta)$: called functions

$$df = \sum_i d\theta_i \frac{\partial f}{\partial \theta_i}$$

e.g. $f(\theta_1, \theta_2) = C_0 + C_1\theta_1 + C_2\theta_2 + C_{12}\theta_1\theta_2$

$$\frac{\partial f}{\partial \theta_2} = C_2 - C_{12}\theta_1$$

$$\frac{\partial \theta_1\theta_2}{\partial \theta_2} = \frac{-\partial \theta_2\theta_1}{\partial \theta_2} = -\theta_1$$

• Since functions of GV are at most linear in each variable, the differential operator $\partial/\partial\theta_i$ is nilpotent : $(\partial/\partial\theta_i)^n = 0$, $n > 1 \in \mathbb{Z}$

• In fact, these operators, together with θ_i's, form a Clifford algebra:

$$\begin{cases} \theta_i\theta_j + \theta_j\theta_i = 0 \\ \frac{\partial}{\partial\theta_i}\frac{\partial}{\partial\theta_j} + \frac{\partial}{\partial\theta_j}\frac{\partial}{\partial\theta_i} = 0 \\ \theta_i \frac{\partial}{\partial\theta_j} + \frac{\partial}{\partial\theta_j}\theta_i = \delta_{ij} \end{cases}$$

- Integration over GV is defined to be identical to differentiation:

$$\int d\theta_i \, f(\theta_1, \dots, \theta_n) = \frac{\partial}{\partial \theta_i} f(\theta_1, \dots, \theta_n)$$

$$\int \theta_i \, d\theta_i = 1 \quad , \quad \int 1 \, d\theta_i = 0$$

Group

non-empty set G, together with a binary operation, $\cdot : G \times G \longrightarrow G$, s.t.
$(a, b) \longmapsto ab$

① $\forall a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

② $\exists ! \, e \in G$, s.t. $ae = ea = a$, $\forall a \in G$

③ $\forall a \in G$, $\exists b$, s.t. $ab = ba = e$, $b \equiv a^{-1}$

Ring

set R, with two binary operations $+$ and $\cdot$,

$+ : R \times R \longrightarrow R$
$(r_1, r_2) \longmapsto r_1 + r_2$, s.t.

$\cdot : R \times R \longrightarrow R$
$(r_1, r_2) \longmapsto r_1 r_2$

① R is an abelian group under addition

$\forall a, b, c \in R$, $(a+b)+c = a+(b+c)$

$\exists 0 \in R$, s.t. $a+0 = 0+a = 0$, $\forall a \in R$

$\forall a \in R$, $\exists b$ s.t. $a+b = 0$, $b \equiv -a$

$a+b = b+a$, $\forall a, b \in R$

② R is associative under multiplication

$(a \cdot b) \cdot c = a \cdot (b \cdot c)$, $\forall a, b, c \in R$

(Not necessary) $\exists 1 \in R$ s.t. $1 \cdot a = a \cdot 1 = a$, $\forall a \in R$

③ Multiplication is distributive w.r.t. addition

$a \cdot (b+c) = ab + ac$, $\forall a, b, c \in R$

$(b+c)a = ba + ca$, $\forall a, b, c \in R$

Field

A commutative ring F, $ab = ba$, s.t. every non-zero element has its multiplication inverse. $\forall a \in F$, $\exists b$ s.t. $ab = ba = 1$.